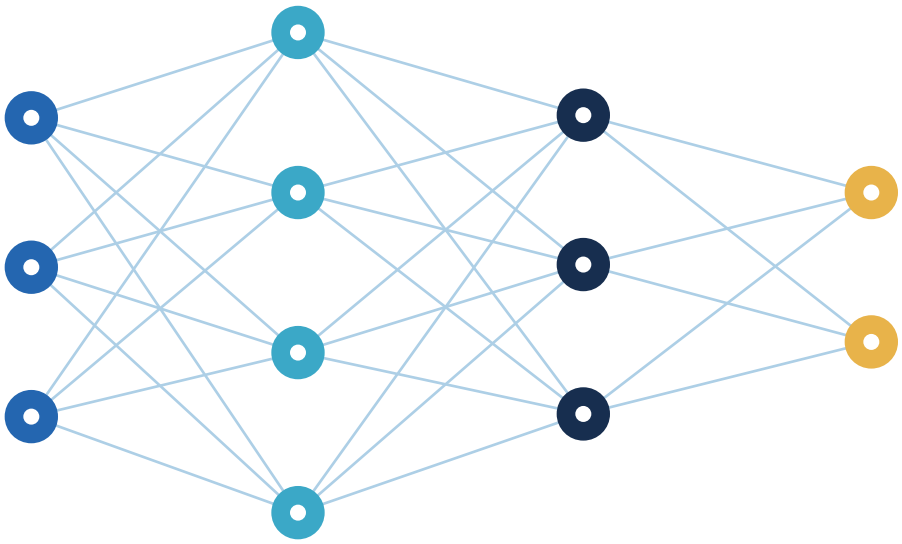
B

BRIGHTLEARN

ADVANCED ANALYTICS MACHINE LEARNING

RESEARCH ASSIGNMENT

Machine Learning Research Assignment



PREPARED B

Toy Mogohlwane

BrightLearn Machine Learning

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Part A - What Is Machine Learning?

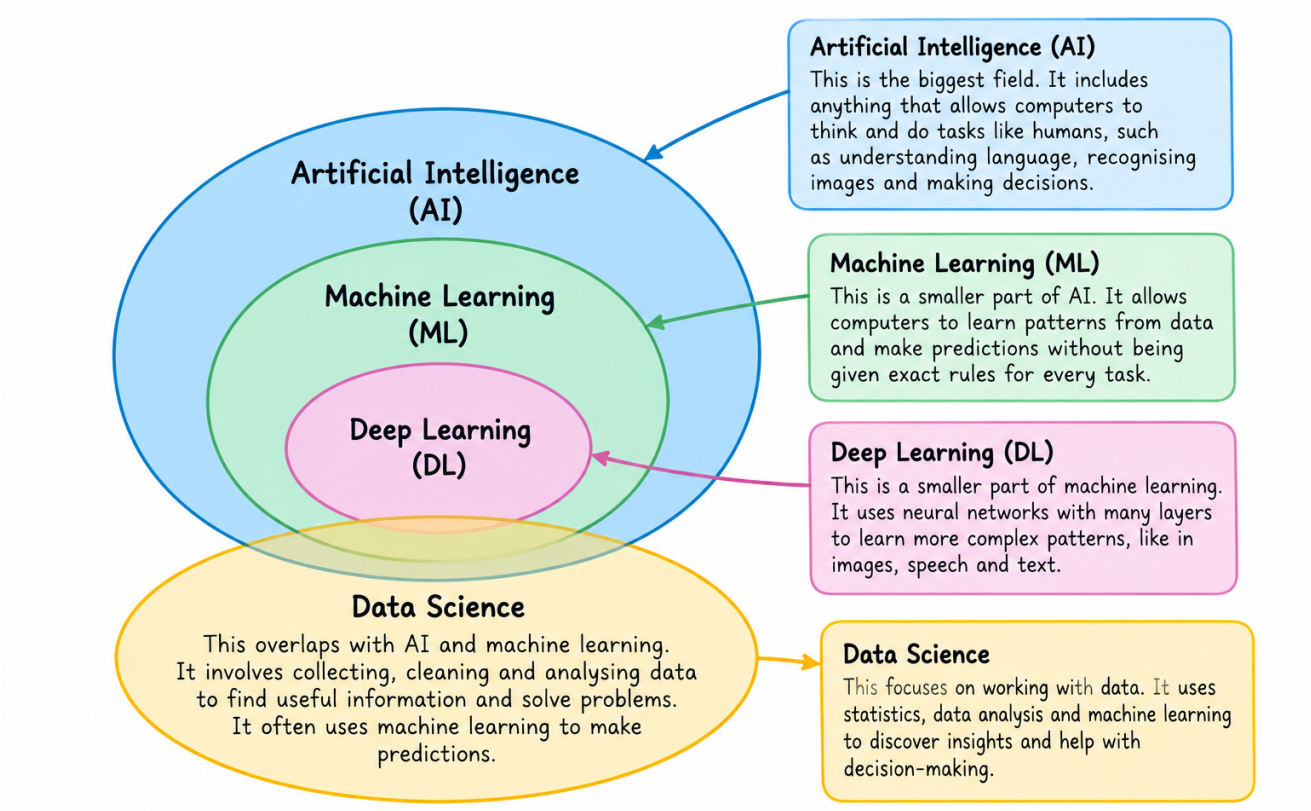
A1. What Is Machine Learning?

Machine learning (ML) is a part of artificial intelligence in which computers find patterns in examples and use them to predict new outcomes. Traditional programming follows rules written by a person, machine learning learns useful rules from data. For example, a programmer might flag emails containing certain words as spam, while an ML model learns from labelled spam and non-spam emails.

Simple comparison: Traditional programming: data + written rules to results. Machine learning: data + known answers to learning algorithm to trained model to predictions.

A2. AI, ML, Deep Learning and Data Science

Artificial intelligence (AI) describes systems performing tasks associated with human intelligence. ML is one approach within AI. Deep learning is a type of ML that uses neural networks with many layers, often for images or speech. Data science collects, cleans, analyses and interprets information, sometimes using ML. Their relationship is AI to ML to deep learning, while data science overlaps with all three.



A3. Core Machine Learning Terms

Using a student exam dataset: a dataset is the table of students, a feature is an input such as study hours or attendance, a label (target) is the exam mark to predict. A model learns the relationship between inputs and marks. Training means learning from previous student records. A prediction is the estimated mark for a new student.

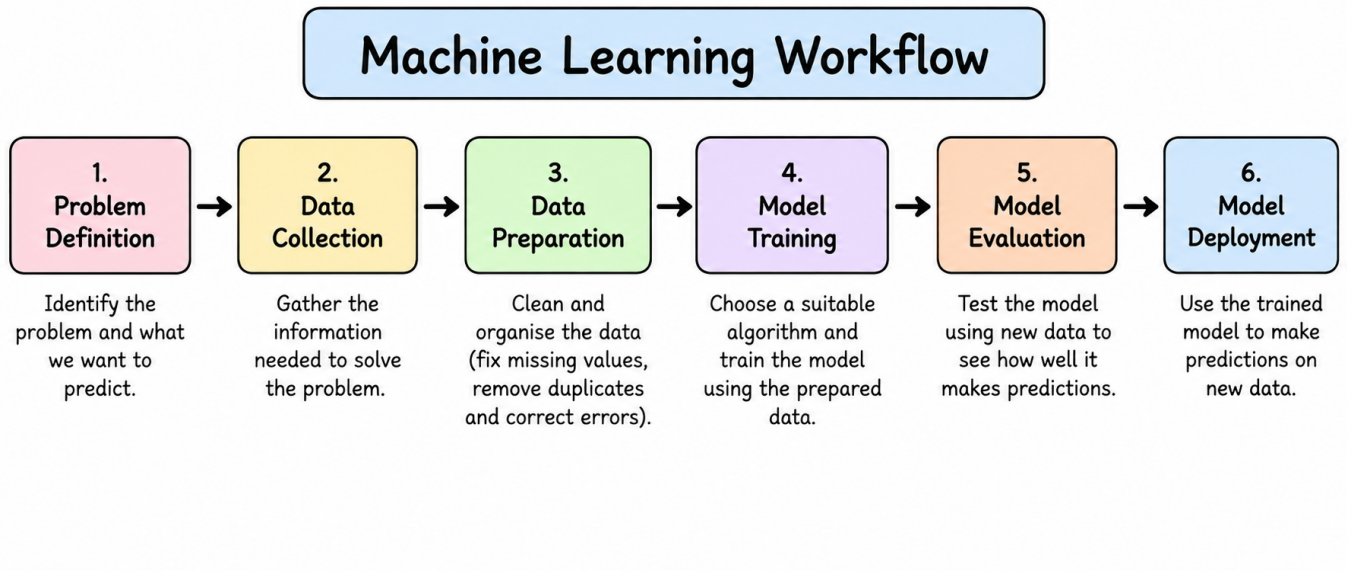
Study hours	Attendance	Exam mark
2	60%	45%
4	75%	58%
6	85%	72%
8	95%	88%

A4. Three African Uses of Machine Learning

Bank fraud detection: Transaction amounts, locations and times help predict suspicious payments, banks and customers benefit. Credit scoring: Income, debt and repayment history help predict repayment risk, lenders and applicants benefit from better decisions. Crop disease detection: Crop images and environmental information help identify disease, farmers benefit through earlier treatment and reduced losses.

A5. Machine Learning Workflow

A typical workflow is: define the problem to collect data to prepare data to select and train a model to evaluate it to deploy it. Data preparation often takes the longest because real datasets contain missing values, duplicates, inconsistent types and errors that must be investigated before training.



Part B - Supervised vs Unsupervised Learning

B1. Supervised Learning

Supervised learning trains on labelled data, where the correct answer is known. For example, a bank can learn from past loan records to predict whether a new customer is likely to repay.

B2. Unsupervised Learning

Unsupervised learning looks for patterns in data without predefined answers. A supermarket might group customers by purchasing habits to understand different shopping behaviours.

B3. Comparison

Aspect	Supervised	Unsupervised
Goal	Predict known targets	Discover patterns
Data	Labelled	Unlabelled
Tasks	Classification, regression	Clustering, PCA
Algorithms	Logistic regression, trees	K-Means, PCA
Evaluation	Accuracy, MAE	Silhouette score (clustering)
Example	Loan repayment prediction	Customer grouping

B4. Classification and Regression

Classification predicts a category, such as fraudulent or legitimate. Regression predicts a number, such as the selling price of a house. Both are supervised because training examples contain the correct outcomes.

B5. Clustering and Dimensionality Reduction

Clustering groups similar records, such as customers with similar spending habits. Dimensionality reduction combines or removes features while preserving useful information, such as simplifying many related patient measurements. Both are unsupervised techniques.

B6. Five Machine Learning Scenarios

(a) House selling price: supervised regression, because the answer is numerical. (b) Customer spending groups: unsupervised clustering, because no groups are predefined. (c) Spam SMS: supervised classification, because messages have category labels. (d) Learner final mark:

supervised regression, because the result is a number. (e) Products with similar sales patterns: unsupervised clustering, because the goal is to discover groups.

B7. Scikit-learn Algorithms

Class / module	Type and use
sklearn.linear_model.LinearRegression	Supervised: predict numbers
sklearn.linear_model.LogisticRegression	Supervised: classify categories
sklearn.tree.DecisionTreeClassifier	Supervised: classify using splits
sklearn.neighbors.KNeighborsClassifier	Supervised: classify from neighbours
sklearn.cluster.KMeans	Unsupervised: cluster records
sklearn.decomposition.PCA	Unsupervised: reduce dimensions

Part C - Preparing the Data

C1. Exploring and Cleaning

First inspect the dataset's shape, types and quality. df.info() shows columns and types, df.describe() summarises numbers, df.isnull().sum() counts missing values. Use df.duplicated() to find repeated rows, df.drop_duplicates() to remove them and df.astype() to correct types.

C2. Missing Values

Use dropna() to remove records with missing data when only a few are affected, on small datasets this can waste valuable observations. fillna() or SimpleImputer can replace values. Mean suits roughly balanced numeric data, median is safer with outliers, mode suits common categories, and a constant such as "Unknown" suits missing labels. For ages 20, 25 and 30, a missing value could be replaced by the mean, 25.

C3. Encoding Categories

Many algorithms need numerical inputs. LabelEncoder gives categories numbers and is mainly suitable for targets. OrdinalEncoder is for ordered values such as low, medium and high. OneHotEncoder or pd.get_dummies() makes separate 0/1 columns for unordered categories such as payment methods. Nominal categories have no natural order, ordinal ones do. One-hot encoding hundreds of unique cities creates too many columns, so rare groups may need combining.

C4. Feature Scaling

Scaling stops large-valued features, such as salary, from dominating distance-based models beside smaller features such as age. StandardScaler uses (value - mean) / standard deviation. MinMaxScaler uses (value - minimum) / (maximum - minimum), usually producing values between 0 and 1. RobustScaler uses the median and interquartile range to reduce outlier influence. KNN, SVM, logistic regression and K-Means often benefit from scaling, decision trees and random forests generally do not need it.

C5. Train/Test Split and Data Leakage

A common split is 80% training and 20% testing, using train_test_split(). The random_state makes results reproducible, stratify keeps class proportions similar. Data leakage occurs when test information influences training. Split first, then fit() imputers, encoders and scalers on training data only. transform() applies learned settings, fit_transform() learns and applies them together. The test data must use transform(), not fit again.

C6. Outliers

Outliers are unusually high or low values. Boxplots display them visually, the IQR rule flags values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR. A z-score beyond about +/-3 may also flag unusual values. An accidental extra zero in a price is an error, but a genuine large purchase is valuable information. Investigate before deleting, df.quantile() and df.clip() can help analyse or cap extremes.

C7. Feature Engineering

Feature engineering creates useful columns from existing ones. In retail sales, Total Amount = Quantity x Price Per Unit, three items at R50 give R150. Another example is extracting the month from a date using pd.to_datetime() and .dt.month. A correlation matrix from df.corr() helps reveal related numeric features, although correlation alone does not prove usefulness or causation.

Part D - Further Machine Learning Research

D1. Overfitting and Underfitting

Overfitting occurs when a model learns training examples too closely, including noise, and performs poorly on new data. A house-price model might memorise training properties rather than learn general price patterns. Underfitting occurs when a model is too simple to capture important relationships, so it performs poorly on both training and test data. For example, predicting house prices from bedroom count alone ignores location and property size.

A train/test split helps reveal the difference: unusually strong training performance with weak test performance suggests overfitting, while weak performance on both suggests underfitting. On a graph, an underfitted curve is too simple for the pattern, an overfitted curve bends around individual observations, and a well-fitted curve follows the main trend.

Overfitting can be reduced with simpler models, regularisation, cross-validation or more representative data. Underfitting may improve with relevant features, a more suitable model or less restrictive settings. The goal is to generalise to unseen examples, not memorise the training set.

Source: Scikit-learn Developers (n.d.), Underfitting vs. overfitting.

D2. Model Evaluation Metrics

Evaluation metrics measure how useful model predictions are. In classification, accuracy is the share of correct predictions. Precision is the proportion of predicted positives that are truly positive, recall is the proportion of actual positives found. F1-score balances precision and recall. A confusion matrix shows true positives, true negatives, false positives and false negatives, helping us see the types of errors a model makes.

Accuracy can be misleading with imbalanced data. If only one of every 100 transactions is fraudulent, a model predicting "not fraud" every time would achieve 99% accuracy while detecting no fraud. Recall is especially useful when missing fraudulent transactions is costly, while precision matters when too many false alarms disrupt legitimate customers. F1-score provides a balance between the two.

Regression uses different measures. Mean Absolute Error (MAE) is the average absolute prediction error, Root Mean Squared Error (RMSE) penalises large errors more strongly, R2 describes how much variation the model explains compared with a baseline. Choose metrics according to the task and the real cost of mistakes.

Source: Scikit-learn Developers (n.d.), Metrics and scoring.

D3. Cross-Validation

Cross-validation estimates how well a model will perform on unseen data by evaluating it across several training and validation splits. In five-fold cross-validation, the available training data is divided into five folds. The model trains five times, each time using four folds for training and the remaining fold for validation. Each fold serves as the validation set once.

The five scores are averaged, and their variation can show whether performance depends heavily on the particular split. This is more reliable than a single validation split because the evaluation uses several different subsets of the data. It is also useful for comparing algorithms and selecting hyperparameters. Scikit-learn provides sklearn.model_selection.cross_val_score() to run this process.

For classification, stratified folds can help preserve class proportions. Preprocessing must happen within each training fold to avoid leakage, so a pipeline is a useful approach. A separate held-out test set should still be kept for the final evaluation after model selection. Cross-validation takes additional computation because the model is trained repeatedly, but it can produce a more dependable estimate of performance.

Source: Scikit-learn Developers (n.d.), Cross-validation: Evaluating estimator performance.

D4. Parameters and Hyperparameters

Parameters are values learned from the training data. For example, linear regression learns coefficients that describe relationships between features and a target. Hyperparameters are settings chosen before training or selected through tuning. They influence how the model learns or how complex it becomes.

In a decision tree, the learned splitting rules are part of the fitted model, while max_depth is a hyperparameter controlling the number of levels. In K-nearest neighbours, the number of neighbours, n_neighbors, is a hyperparameter. If the settings are too flexible, a model may overfit, if they are too restrictive, it may underfit.

Scikit-learn's GridSearchCV tries combinations of specified hyperparameter values and evaluates them using cross-validation. It selects the combination that performs best according to a chosen score. However, it can be slow because each combination requires repeated model training. For example, testing ten values for two settings across five folds can require hundreds of fits. Randomized search is one alternative when the search space is large. Understanding both types of values helps us improve performance systematically.

Source: Scikit-learn Developers (n.d.), Tuning the hyper-parameters of an estimator.

D9. Machine Learning Pipelines

A machine learning pipeline connects preprocessing and model training into one repeatable workflow. In scikit-learn, sklearn.pipeline.Pipeline runs steps in order, such as imputing missing values, scaling features and fitting a model. sklearn.compose.ColumnTransformer applies different transformations to different columns, for example scaling numeric columns while one-hot encoding payment methods.

In my Shop Performance practical, I filled missing Quantity values, handled Discount and PaymentMethod, encoded payment categories and scaled Quantity and Discount. These steps were performed separately, but they could be combined into a pipeline. A pipeline learns preprocessing settings from training data and applies the same settings to test data, reducing the chance of data leakage or inconsistent transformations.

Pipelines also reduce repeated code, make results easier to reproduce and work well with cross-validation or hyperparameter searches. When the pipeline is evaluated inside cross-validation, the imputer and scaler are fitted again on each training fold rather than on the validation fold. This is important because even a simple average calculated using test

information can make evaluation misleading. A well-designed pipeline makes a machine learning project easier to maintain and apply to new records.

Source: Scikit-learn Developers (n.d.), Pipeline: Chaining estimators.

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AI-Use Declaration

I used ChatGPT as a support tool while completing my Machine Learning Research Assignment. It helped me understand machine learning concepts, clarify explanations and organise my research. It also provided guidance on Python coding and data preprocessing.

I used Databricks to run Python code, examine the results and complete the practical section. I used Canva to prepare my original research report. AI was used as a learning and guidance tool. Tool used: ChatGPT (OpenAI). Purpose: learning support, research assistance and coding guidance.